Check of `Evolution_terms.tex`: the evolution terms, JIMWLK, BFKL and their colour-singlet forms

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1 Summary

Throughout, $\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$, the prefactor of every equation of your file. JIMWLK⊗LO and BFKL⊗LO are the targets of the earlier notes (Eqs. (3.1), (3.2) of `Evolution_singlet.pdf`).

What is correct.

1. **Sec. 1, genuine terms.** All 8 equations agree with the checked genuine terms (Group I Rows II, V; Group II Rows I, IV; Group III Row III), Table 1.
2. **Sec. 2, from three and four ρ's.** All 15 equations are identical, term by term, to the lists of the previous version, which were checked against the reordered three- and four-ρ terms (Table 2).
3. **Sec. 3, JIMWLK.** Your collected form, Eq. (3.3) of your file, *is* JIMWLK⊗LO, exactly, for any adjoint U (3.4×10^{-17}). $dN_1 + dN_2$ equal it with your convention $U^{bc} = U^{cb}$.
4. **Sec. 4, BFKL.** Your collected form, Eq. (4.2), *is* BFKL⊗LO, exactly, for any U (8.7×10^{-16}). Eq. (4.1) equals it with $U^{bc} = U^{cb}$.
5. **Colour-singlet forms** (Eqs. (3.4), (4.3)). Both are correct for any U:
   - they equal the singlet projection of your own collected forms (2.3×10^{-15}, 6.0×10^{-16});
   - they equal Eqs. (3.1), (3.2) of `Evolution_singlet.pdf` (1.4×10^{-15}, 4.2×10^{-16}).

What is not.

P1 Secs. 1 and 2 do not add up to Sec. 3 (this is M1 of `Final_evolution_check.pdf`, still open). The "From 4 ρ's" list of Sec. 2 is the one of the previous version. The two-ρ lists of `4rho.tex` (30 terms) are still missing.

- Without them, Secs. 1+2 differ from JIMWLK⊗LO by 8–54% (relative, at random points, your convention), and by 27–170% in the singlet projection.
- The differing kernel structures are $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$ times $\frac{1}{(x-z)^2}$, $\frac{1}{(x'-z)^2}$ and $K(x,x',z)$.
- With them, Secs. 1+2 equal Sec. 3 exactly (8.4×10^{-17}). The 30 terms are written out in Eq. (4.1).

P2 The BFKL section: which logarithm.

- As written, both Sec. 3 and Sec. 4 are $d^3N_{\rm NLO}/d^3k \propto \log\frac{\vee}{\Lambda}$. Added, they would give $\log\frac{\vee}{\Lambda}\,(\mathcal{J}+\mathcal{B})$: the evolution twice.

- The $\log\frac{\vee}{\Lambda}$ terms of Secs. 1, 2 sum to $\mathcal{J}$ alone (Sec. 3, P1 fixed). None of them has a BFKL kernel: in all 504 the measured gluon is emitted by a valence charge, while in 23 of the 35 terms of Eq. (4.2) it is emitted by the gluon at z.
- $\mathcal{B}$ comes from the upper limits of the p^+ integrals, with $\log\frac{\vee}{k^+}$: the $\log\frac{\vee}{k^+}$ terms of the rows sum to $\log\frac{\vee}{k^+}(\mathcal{B}-\mathcal{J})$.
- At mid rapidity both evolutions take half of the window: $\frac{1}{2}\log\frac{\vee}{\Lambda}\,\mathcal{J}+\frac{1}{2}\log\frac{\vee}{\Lambda}\,\mathcal{B}$, and the finite remainder is $\log\frac{k^+}{\sqrt{\vee\Lambda}}\,(\mathcal{J}-\mathcal{B})$, Eq. (5.3).
- In Sec. 4, $\log\frac{\vee}{\Lambda}\to\frac{1}{2}\log\frac{\vee}{\Lambda}$; in the final result Sec. 3 also enters with $\frac{1}{2}\log\frac{\vee}{\Lambda}$ (Sec. 5).

P3 Normalization of the singlet forms. The bracket is right but the prefactor is not complete (Sec. 6):

- $\frac{1}{N_c^2-1}\langle\rho^f(x)\rho^f(y)\rangle$ is missing;
- the integral is written over x', the integrand uses y;
- $K(a,b,z)$ is used before it is defined, and only $K(y,y,z)$ and $K(x,y,z)$ are defined.

P4 Labels (Sec. 7).

- In Group I Row V, the fourth equation is labelled $d^3N_{2b\,\text{finite}}$ a second time. It is the counterterm: it equals $-d^3N_{2b\,\text{UV}}$ term by term, so the two cancel (4.9×10^{-17}).
- "(with c.c.)" in the headings means that the c.c. is already written out. Adding it once more would be wrong: for Group III Row III it would change the result by 8.3×10^{-2}.

The convention $U^{bc}=U^{cb}$. Some of your forms are correct only with your convention $U^{bc}=U^{cb}$ ($U^\dagger=U$), as found in `Evolution_singlet.pdf`:

- the Sec. 2 lists;
- the genuine Group II Row IV and Group III Row III;
- dN_1+dN_2 of Sec. 3;
- Eq. (4.1).

The collected forms (3.3), (4.2) and the singlet forms (3.4), (4.3) are correct for any U. Nothing has to be changed for this, as long as the convention is stated.

Corrected copy. `Evolution_terms_corrected.tex` (and `.pdf`) is your file with P1–P4 fixed, removed text in red, new text in blue. Parsed back, its Secs. 1+2 equal its Sec. 3 and JIMWLK⊗LO (8.4×10^{-17}).

2 How the check was done

- Every term of the file was read by a parser: 624 two-ρ terms in 30 equations, and the 44 + 8 trace monomials of the two singlet forms. The factored forms "kernel $\times\,[\,\dots\,]$" were expanded.
- Each term was converted to the format of the earlier checks (Wilson lines, structure constants, charges, kernels).
- Two sums of terms are compared at random transverse points x,x',w,w',z, with random classical charges $\rho^a(x)$, $\rho^a(x')$ and random SU(3) adjoint Wilson lines. Charges are not projected, so this is stronger than the singlet comparison.
  - The number given is $|A-B|$ divided by the sum of the moduli of all single terms, maximized over three configurations.
  - Agreement means $\lesssim10^{-15}$.
  - "$U=U^T$" is your convention $U^{bc}=U^{cb}$; "any U" is a general adjoint U.
- Singlet forms: the traces are evaluated with the same random U, with $x\to p_1$, $y\to p_2$, symmetrized.

3 Equation by equation

Table 1: Sec. 1, genuine terms, against the checked genuine terms of `Final_evolution_check.pdf`.

equation	$U = U^T$	any U	
(1.1) Group I, Row II	3.0×10^{-17}	2.1×10^{-17}	agrees, any U
(1.2) Group I, Row V, $d^3 N_{2a}$	6.0×10^{-17}	7.9×10^{-17}	agrees, any U
(1.3) Group I, Row V, $d^3 N_{2b\,\mathrm{UV}}$	3.6×10^{-17}	2.1×10^{-17}	agrees, any U
(1.4) Group I, Row V, $d^3 N_{2b\,\mathrm{finite}}$	2.1×10^{-17}	1.3×10^{-16}	agrees, any U
(1.5) Group I, Row V, second "$d^3 N_{2b\,\mathrm{finite}}$" vs counterterm	3.6×10^{-17}	2.1×10^{-17}	agrees, any U
(1.3) + (1.5), relative to (1.3) (vanishes)	4.9×10^{-17}	5.9×10^{-17}	agrees, any U
(1.2)–(1.5) together vs Row V	2.5×10^{-17}	3.1×10^{-17}	agrees, any U
(1.6) Group II, Row I	3.4×10^{-17}	9.7×10^{-18}	agrees, any U
(1.7) Group II, Row IV	2.0×10^{-16}	3.0×10^{-1}	agrees for $U^{bc} = U^{cb}$
(1.8) Group III, Row III, second part (as written)	1.3×10^{-16}	2.1×10^{-1}	agrees for $U^{bc} = U^{cb}$
(1.8) with its c.c. added once more	8.3×10^{-2}	3.1×10^{-1}	differs
all of Sec. 1	7.1×10^{-17}	5.3×10^{-2}	agrees for $U^{bc} = U^{cb}$

Table 2: Sec. 2 against the same lists of the previous version, which were checked against the reordered three- and four-ρ terms.

equation	terms (new / previous)	$U = U^T$	any U	
(2.1) From four ρ's	20 / 20	0	0	identical
(2.2) Group I, Row II	22 / 22	0	0	identical
(2.3) Group I, Row III, N_1	21 / 21	0	0	identical
(2.4) Group I, Row III, N_2	24 / 24	0	0	identical
(2.5) Group I, Row IV, N_1	33 / 33	0	0	identical
(2.6) Group I, Row IV, N_2	13 / 13	0	0	identical
(2.7) Group I, Row V	16 / 16	0	0	identical
(2.8) Group II, Row I	31 / 31	0	0	identical
(2.9) Group III, Row I, N_1	30 / 30	0	0	identical
(2.10) Group III, Row I, N_2	16 / 16	0	0	identical
(2.11) Group III, Row II, N_1	34 / 34	0	0	identical
(2.12) Group III, Row II, N_2	11 / 11	0	0	identical
(2.13) Group III, Row III	16 / 16	0	0	identical
(2.14) Group III, Row V	24 / 24	0	0	identical
(2.15) Group III, Row VI	24 / 24	0	0	identical

Table 3: Secs. 3, 4 and the totals (normalized by the sum of the moduli of the single terms, Sec. 2).

comparison	$U = U^T$	any U	
Secs. 1+2 vs JIMWLK⊗LO	2.4×10^{-2}	1.8×10^{-2}	differs
Secs. 1+2 + `4rho.tex` lists (P1) vs JIMWLK⊗LO	8.4×10^{-17}	5.2×10^{-2}	correct for $U^{bc} = U^{cb}$
(3.1) + (3.2) vs (3.3)	7.7×10^{-17}	1.1×10^{-1}	agrees for $U^{bc} = U^{cb}$
(3.3) vs JIMWLK⊗LO	1.5×10^{-16}	3.4×10^{-17}	correct, any U
(3.3) vs Secs. 1+2 as written	2.5×10^{-2}	1.8×10^{-2}	differs
(4.1) vs (4.2)	4.6×10^{-17}	1.3×10^{-1}	agrees for $U^{bc} = U^{cb}$
(4.2) vs BFKL⊗LO	2.2×10^{-15}	8.7×10^{-16}	correct, any U
(4.1) vs BFKL⊗LO	2.2×10^{-15}	9.2×10^{-2}	correct for $U^{bc} = U^{cb}$

Table 4: The colour-singlet forms (relative difference of the values).

comparison	$U = U^T$	any U	
(3.4) vs singlet projection of (3.3)	1.0×10^{-15}	2.3×10^{-15}	agrees, any U
(3.4) vs Eq. (3.1) of `Evolution_singlet.pdf`	1.6×10^{-15}	1.4×10^{-15}	correct, any U
(4.3) vs singlet projection of (4.2)	1.1×10^{-15}	6.0×10^{-16}	agrees, any U
(4.3) vs Eq. (3.2) of `Evolution_singlet.pdf`	9.3×10^{-16}	4.2×10^{-16}	correct, any U

4 P1: the missing two-ρ terms of `4rho.tex`

These are the two-ρ terms of `4rho.tex` ("Two ρ contribution", $dN^{(1)}$, $dN^{(2)}$ with c.c.), with the fixes F1, F2 of `Final_evolution_check.pdf`:

- **F1:** in the first part, drop $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$;
- **F2:** in the second part, $dN^{(1)}$, term 2 has one index four times. The correct term is

$$3f^{abc}f^{deg}\rho^{b'}(x')\rho^{c}(y)U^{ag}(y)U^{db'}(w')U^{eb}(z).$$

They belong to "From 4 ρ's" of Sec. 2. Collected, in your notation (to be added):

$$\left.\frac{d^3N_{\rm NLO}}{d^3k}\right|_{\texttt{4rho.tex}} = \mathcal{N}\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[$$

$$\begin{aligned}
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^l(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^e(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x')\Bigg]
\end{aligned}\tag{4.1}$$

This list, read back by the parser, equals the checked `4rho.tex` lists (1.2×10^{-16} for $U=U^T$).

5 P2: which logarithm multiplies BFKL

Every row is an integral over the plus momentum p^+ of the unobserved gluon, with limits built from Λ, k^+ and $\vee$:

- the lower end, $p^+ \to \Lambda$ (the unobserved gluon softer than the measured one), gives $\log \frac{k^+}{\Lambda}$; summed over all rows, its coefficient is JIMWLK⊗LO, $\mathcal{J}$;
- the upper end, $p^+ \to \vee$ or $p^+ + k^+ \to \vee$ (the unobserved gluon harder), gives $\log \frac{\vee}{k^+}$; summed over all rows, its coefficient is BFKL⊗LO, $\mathcal{B}$ (earlier notes).

So the large logarithms of the NLO result are, exactly,

$$\frac{d^3N_{\rm LL}}{d^3k} = \mathcal{N}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}\Big]. \tag{5.1}$$

With your rule A, $\log\frac{k^+}{\Lambda} = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+}$, this is how your rows give it:

$$\frac{d^3N_{\rm LL}}{d^3k} = \mathcal{N}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\log\frac{\vee}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,(\mathcal{B}-\mathcal{J})\Big]. \tag{5.2}$$

- $\mathcal{J}$ is the bracket of your Sec. 3: the coefficient of $\log\frac{\vee}{\Lambda}$, i.e. the sum of all terms of Secs. 1, 2 (with P1).
- $\mathcal{B}$ is the bracket of your Sec. 4: the coefficient of $\log\vee$. It is not among the $\log\frac{\vee}{\Lambda}$ terms.

The finite logarithm is the distance in rapidity of the measured gluon from the centre of the window, $\Delta \equiv \log\frac{k^+}{\sqrt{\vee\Lambda}}$, so that $\log\frac{\vee}{k^+} = \frac{1}{2}\log\frac{\vee}{\Lambda} - \Delta$:

$$\boxed{\frac{d^3N_{\rm LL}}{d^3k} = \mathcal{N}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\frac{1}{2}\log\frac{\vee}{\Lambda}\,\mathcal{J} + \frac{1}{2}\log\frac{\vee}{\Lambda}\,\mathcal{B} + \log\frac{k^+}{\sqrt{\vee\Lambda}}\,(\mathcal{J}-\mathcal{B})\Big]} \tag{5.3}$$

- $\frac{1}{2}\log\frac{\vee}{\Lambda}\,\mathcal{J}$: JIMWLK evolution of the target (your Sec. 3 with $\frac{1}{2}$);
- $\frac{1}{2}\log\frac{\vee}{\Lambda}\,\mathcal{B}$: BFKL evolution of the projectile (your Sec. 4 with $\frac{1}{2}$);
- $\log\frac{k^+}{\sqrt{\vee\Lambda}}\,(\mathcal{J}-\mathcal{B})$: finite; it vanishes at mid rapidity, $k^+ = \sqrt{\vee\Lambda}$.

Counting $\log\frac{\vee}{k^+}$ itself as finite would leave BFKL inside the finite part, $\log\frac{\vee}{k^+}(\mathcal{B}-\mathcal{J})$, which at mid rapidity is $\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{B}-\mathcal{J})$, not finite.

In your Sec. 4 (all three equations): $\log\Big[\frac{\vee}{\Lambda}\Big] \to \frac{1}{2}\log\Big[\frac{\vee}{\Lambda}\Big]$, and the total (5.3) is to be stated; otherwise Sec. 3 + Sec. 4 count $\log\frac{\vee}{\Lambda}$ twice.

6 P3: the prefactor of the singlet forms

With the projectile in a colour singlet, $\rho^b(x')\rho^c(x) \to \delta^{bc}\langle\rho^f(x)\rho^f(y)\rangle/(N_c^2-1)$ (then $x' \to y$), your Eq. (3.4) reads

$$\frac{d^3N_{\rm NLO}}{d^3k} = \mathcal{N}\log\Big[\frac{\vee}{\Lambda}\Big]\frac{1}{N_c^2-1}\int_{x,x'y,z,w,w'} e^{-ik\cdot(w'-w)}\,\langle\rho^f(x)\rho^f(y)\rangle\Big[\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[\dots\Big]+\dots\Big], \tag{6.1}$$

and the same for Eq. (4.3), with $\frac{1}{2}\log\Big[\frac{\vee}{\Lambda}\Big]$ (P2). Define once, before Eq. (3.4):

$$K(a,b,z) \equiv \mathcal{K}(a-z)\cdot\mathcal{K}(b-z) = \frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}, \qquad K(a,a,z) = \frac{1}{(a-z)^2}. \tag{6.2}$$

With this, $\mathcal{N}\log\frac{\vee}{\Lambda}/(N_c^2-1) = P/k^+$ with $P = \frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}$ of your App. F.

7 P4: labels

- **Group I Row V.** The four lists are d^3N_{2a}, $d^3N_{2b\,\rm UV}$, $d^3N_{2b\,\rm finite}$ and a fourth one, also called $d^3N_{2b\,\rm finite}$.
  - The fourth is $-d^3N_{2b\,\rm UV}$, term by term: $d^3N_{2b\,\rm finite} \to d^3N_{2b\,\rm CT}$ (counterterm).
  - The sum of all four is the checked Row V (Table 1).
- **"(with c.c.)".**
  - In Sec. 1 the heading of Group I no longer says "(with c.c.)"; the heading of Group III does.
  - In both groups the c.c. is already written out in the lists, as it is in all Sec. 2 lists with this heading.
  - A note such as "c.c. included" would avoid adding it twice.

Files

In `nlo_gluon_paper/check_v4/`:

- `parse_v4.py`: parser of all terms and of the trace forms;
- `check_all.py`: Tables 1–3;
- `check_singlet.py`: Table 4;
- `check_m1.py`, `m1_singlet.py`: P1;
- `make_corrected.py`: the corrected copy, parsed back;
- `make_check.py`: this note.